

README for replication archive of “Real Effects of Markets on Politics: Evidence from U.S. Presidential Elections,” by Alan Crane, Andy Koch, and Leming Lin.

Overview

The code in this replication package constructs the analysis presented in “Real Effects of Markets on Politics: Evidence from U.S. Presidential Elections,” by Alan Crane, Andy Koch, and Leming Lin. The included data and programs replicate the 1 figure and 3 tables from the paper as well as the appendix tables and figures. The analysis uses Stata and SAS.

Data Availability and Provenance Statements

Statement about Rights

- X I certify that the author(s) of the manuscript have legitimate access to and permission to use the data used in this manuscript.
- X I certify that the author(s) of the manuscript have documented permission to redistribute/publish the data contained within this replication package.

Summary of Availability

- ☐ All data **are** publicly available.
- X Some data **cannot be made** publicly available.
Note: Analysis can be replicated from included processed intermediate data files, but certain raw data files cannot be included due to licensing restrictions.
- ☐ **No data can be made** publicly available.

Details on each Data Source

Below is a table with all data sources used in the paper and internet appendix. Each data source is matched to the corresponding data files included in the archive and an indicator if those data are included in the archive or are excluded due to licensing restrictions. ***Sample versions of files associated with restricted sources are included so the data fields and types can be replicated from the vendors.***

Data Source Summary Table

Data.Name	Data.Files	Provided	Citation
“Presidential Election Voting Outcomes By County”	cq_election.xlsx vote_cny.sas	FALSE TRUE	CQ Press (2022)
“IRS Dividend Income Data”	irs_cnty_8916_raw.sas Incyallagi.sas zip_agi0409.sas	TRUE	IRS (2021)
“Employment & Wage By Industry &/or County”	Allhlcn.sas qcew_8820.sas 1990.annual.singlefile.csv ... 2019.annual.singlefile.csv	TRUE	Bureau of Labor Statistics (2022b)
“County Income and Population”	county_income.sas	TRUE	Bureau of Economic Analysis (2022)

Data.Name	Data.Files	Provided	Citation
“County Unemployment”	cnty_unem.sas	TRUE	Bureau of Labor Statistics (2022a)
“County Age and Race”	stch1990.sas ... stch1999.sas coest00intalldata.sas CC-EST2020-ALLDATA.xlsx	TRUE	Census Bureau (2022b)
“Urban & Rural Population”	urban1990_CDC.xlsx PctUrbalRural_County_2010.xlsx	TRUE	Census Bureau (2010b)
“County Education Levels”	Education_attainment_ usda_90_20_145yearACS.xlsx Education_attainment_ACS _2010_5year_estimate.xlsx	TRUE	Census Bureau (2010a)
“GDP Growth”	quarterly_real_GDP_growth.sas	TRUE	Bureau of Economic Analysis (2020)
“Aggregate Wages”	wage_aggregate.sas	TRUE	Bureau of Labor Statistics (2020b)
“Aggregate Unemployment”	unrate_aggregate_monthly.sas	TRUE	Bureau of Labor Statistics (2020c)
“Federal Funds Rate”	fed_fund.sas	TRUE	Board of Governors of the Federal Reserve System (2020)
“Credit Spread”	credit_spread_moody.sas	TRUE	Moody’s (2020)
“Monthly CPI”	cpi.monthly.sas	TRUE	Bureau of Labor Statistics (2020a)
“NAICS to SIC Crosswalk”	2002_NAICS_to_1987_SIC.xlsx	TRUE	Census Bureau (2022a)
“Zip to County Crosswalk”	zip_county.sas	TRUE	HUD (2022)
“Value-Weighted Aggregate Stock Returns”	msi.sas dsi.sas	FALSE	Center for Research in Security Prices (2021)
“Corporate Headquarters”	funda_head.sas hq_10k.sas	FALSE TRUE	S&P Global (2020) SEC (2020)

Data Source Item Detail

“IRS Dividend Income Data by county”: data were downloaded from IRS at <https://www.irs.gov/statistics/soi-tax-stats-county-data> (accessed April 2021). Data from 1989 to 2010 are available under “County Income Data 1989–2010”. Data after 2010 are available under “County, Metropolitan, and Micropolitan Data”. A copy of the data is provided as part of this archive. The data are in the public domain. `irs_cnty_8916_raw.sas` – contains dividend and income data from the IRS. `incyallagi.sas` contains more detailed data at the county level since 2010. `zip_agi0409.sas` contains zip code level taxable income data from 2004 to 2009, available at <https://www.irs.gov/statistics/soi-tax-stats-individual-income-tax-statistics-zip-code-data-soi> (accessed April 2021).

Datafiles: `irs_cnty_8916_raw.sas`
 `incyallagi.sas`
 `zip_agi0409.sas`

“Employment & Wage By Industry &/or County”: data were downloaded from the BLS. A copy of the data is provided as part of this archive. The data are in the public domain. `Allhlcn.sas` is from the Quarterly Census of Employment and Wages (QCEW) available at <https://www.bls.gov/cew/downloadable-data-files.htm> QCEW NAICS-Based Data Files: “County High-level” files. The 1990-2019 single files: employment data by county and 4-digit industry from BLS QCEW <https://www.bls.gov/cew/downloadable-data-files.htm> QCEW NAICS-Based Data Files: “Quarterly single files”. `qcew_8820.sas` contains average weekly wage during the last quarter of each year from 1988 to 2020 available at <https://www.bls.gov/cew/downloadable-data-files.htm>. Data from 1990 to 2020 are from QCEW NAICS-Based Data Files: “County High-level” files. Data from 1988 to 1989 are from QCEW SIC-Based Data Files: “CSVs Single Files”. Wage data are for both private and public sector employees.

Datafiles: `Allhlcn.sas`
 `qcew_8820.sas`
 `1990.annual.singlefile.csv`
 `...`
 `2019.annual.singlefile.csv`

“County Income and Population”: data were downloaded from the BEA. The data are in the public domain and a copy of the data is provided as part of this archive as `county_income.sas`, which contains county income per capita and population data. Available on the BEA website, <https://www.bea.gov/itable/regional-gdp-and-personal-income>, Interactive Data Tables: Personal income and employment by county and metropolitan area: County and MSA personal income summary: personal income, population, per capita personal income (accessed September 2022).

Datafiles: `county_income.sas`

“County Unemployment”: data were downloaded from the BLS. The data are in the public domain and a copy of the data is provided as part of this archive as `cnty_unem.sas` which contains county level annual average unemployment rate 1990-2020. Available on the BLS website, <https://www.bls.gov/lau/tables.htm> County Tables: Labor force data by county 1990 - 2020 (accessed September 2022).

Datafiles: `cnty_unem.sas`

“County Age and Race”: data are from the Census Bureau. The data are in the public domain and a copy of the data is provided as part of this archive. `stch1990.sas-stch1999.sas` contain county race and age data from 1990 to 1999 and `coest00intalldata.sas` from 2000 and 2010 from the NBER Public Use Data Archive

<https://www.nber.org/data/census-intercensal-county-population-age-sex-race-hispanic.html>.
Data from 2011-2020 is in CC-EST2020-ALLDATA.sas from the Census
<https://www2.census.gov/programs-surveys/popest/datasets/2010-2020/counties/asrh/>.

Datafiles: stch1990.sas-stch1999.sas
coest00intalldata.sas
C-EST2020-ALLDATA.sas

“Urban & Rural Population”: data are from the Census Bureau. The data are in the public domain and a copy of the data is provided as part of this archive. Data in 1990 are in urban1990_CDC.xlsx and were obtained through the CDC
<https://www2.cdc.gov/nceh/lead/census90/house11/download.htm> Dataset used: cnty.dbf (accessed September 2017). Data in 2010 are in PctUrbalRural_County_2010.xlsx and are obtained from the Census website, <https://www.census.gov/programs-surveys/geography/guidance/geo-areas/urban-rural/2010-urban-rural.html> Dataset used: Percent Urban and Rural In 2010 by State and County (accessed January 2021).

Datafiles: urban1990_CDC.xlsx
PctUrbalRural_County_2010.xlsx

“County Education Levels”: data are from the Census Bureau. The data are in the public domain and a copy of the data is provided as part of this archive. Data in 1990 are in Education_attainment_usda_90_20_145yearACS.xlsx and were obtained through the USDA <https://www.ers.usda.gov/data-products/county-level-data-sets/county-level-data-sets-download-data/> Dataset used: Educational attainment for adults age 25 and older for the U.S., States, and counties, 1970–2020 (accessed January 2021). Variable used: “Percent of adults completing some college or associate’s degree, 1990.” Data in 2010 are in Education_attainment_ACS_2010_5year_estimate.xlsx and were obtained from the 2010 American Community Survey 5-year estimates, <https://data.census.gov/all?q=education> Under Tables S1501 - 2010: ACS 5-Year Estimates Subject Tables; Geography: All counties within United States and Puerto Rico (accessed January 2021). Variable used: “Percent bachelor’s degree or higher” for “population 25 years and over”.

Datafiles: Education_attainment_usda_90_20_145yearACS.xlsx
Education_attainment_ACS_2010_5year_estimate.xlsx

Macroeconomic series: GDP Growth, Aggregate Wages, Aggregate Unemployment, Federal Funds Rate, and Credit Spread are obtained from the Federal Reserve Economic Data (FRED) database a <https://fred.stlouisfed.org/>. The data are in the public domain and a copy of the data is provided as part of this archive combined into macro_vars_all.dta. Separate .sas files are included for each series along with the raw .xls files from the FRED server.

“GDP Growth” is real gross domestic product [a191rl1q225sbea], retrieved from FRED at <https://fred.stlouisfed.org/series/A191RL1Q225SBEA> (accessed December 2022). Data are in quarterly_real_GDP_growth.sas or GDPGROWTH.XLS.

“Aggregate Wages” is retrieved from FRED at <https://fred.stlouisfed.org/series/LES1252881600Q> (accessed December 2022). Data are in wage_aggregate.sas or WAGES.XLS

“Aggregate Unemployment” is retrieved from FRED at <https://fred.stlouisfed.org/series/UNRATE> (accessed December 2022). Data are in unrate_aggregate_monthly.sas or UNRATE.XLS.

“Federal Funds Rate” is retrieved from FRED at <https://fred.stlouisfed.org/series/FEDFUNDS> (accessed December 2022). Data are in fed_fund.sas or FEDFUNDS.XLS.

“Credit Spread” is retrieved from FRED at <https://fred.stlouisfed.org/series/BAA> and <https://fred.stlouisfed.org/series/AAA> (accessed December 2022). Data are in credit_spread_moody.sas or AAA.XLS and BBB.XLS.

Datafiles: macro_vars_all.dta
quarterly_real_GDP_growth.sas or GDPGROWTH.XLS
wage_aggregate.sas or WAGES.XLS
unrate_aggregate_monthly.sas or UNRATE.XLS.
fed_fund.sas or FEDFUNDS.XLS.
credit_spread_moody.sas or AAA.XLS and BBB.XLS.

“Monthly CPI”: data are from the BLS. Consumer price index for all urban consumers: All items in U.S. city average [cpiaucsl]. The data are in the public domain and a copy of the data is provided as part of this archive. Data are in `cpi_monthly.sas` and were retrieved from FRED, Federal Reserve Bank of St. Louis; <https://fred.stlouisfed.org/series/CPIAUCSL> (accessed April 2021).

Datafiles: `cpi_monthly.sas`

Data Crosswalk files: NAICS to SIC and Zip Code to County. Obtained from the Census Bureau and HUD, respectively. The data are in the public domain and a copy of the data is provided as part of this archive.

“NAICS to SIC” is in `2002_NAICS_to_1987_SIC.xlsx` and was retrieved from <https://www.census.gov/naics/?68967> from the dataset “2002 NAICS to 1987 SIC” (Accessed December 2022).

“Zip Code to County” is the USPS zip code crosswalk files retrieved from https://www.huduser.gov/portal/datasets/usps_crosswalk.html. Data year and quarter: 2010 Q1 (Accessed June 2020) and found in archive file `zip_county.sas`.

Datafiles: `2002_NAICS_to_1987_SIC.xlsx`
`zip_county.sas`

“Company Headquarters”: Firm headquarters are collected from two sources. The first is from Compustat accessed via the Wharton Research Data Service (WRDS) <https://wrds-www.wharton.upenn.edu/pages/about/data-vendors/sp-global-market-intelligence/> (accessed December 2020). These data include a company’s most recent headquarters location as of their latest filing. Data are subject to redistribution restriction so the raw data are not included in the archive, but can be easily downloaded with a license. These data should be saved with file name `funda_head.sas` for purpose of replication. Additional headquarters location was collected via 10-K filings collected from the SEC’s EDGAR server at <https://www.sec.gov/edgar/searchedgar/companysearch> (accessed December 2020). These data are in the public domain and included in the archive in file `hq_10k.sas`.

Datafiles: `funda_head.sas`
`hq_10k.sas`

“Value-Weighted Aggregate Stock Returns”: Monthly and daily stock value-weighted aggregate stock returns were downloaded from the Center for Research in Security Prices (CRSP) through the Wharton Research Data Service (WRDS) available at <https://wrds-www.wharton.upenn.edu/pages/about/data-vendors/center-for-research-in-security-prices-crsp/>. These data are subject to redistribution restriction so the raw data are not included in the archive, but can be easily downloaded with a license. These data should be saved with file name `msi.sas` (monthly returns) and `dsi.sas` (daily returns) for purpose of replication.

Datafiles: `msi.sas`
`dsi.sas`

“Presidential Election Voting Outcomes By County”: Elections returns by county for each presidential election from 1980-2020 are downloaded from CQ Voting and Elections Collections available at <https://library.cqpress.com/elections/>. These data are subject to redistribution restriction so the raw data are not included in the archive, but can be easily downloaded with a license. These data should be saved with file name `cq_election.xlsx` for purpose of replication.

Author created measures of vote share generated from the raw data are included for replication purposes in `vote_cny.sas`.

Datafiles: `cq_election.xlsx`
`vote_cny.sas`

Repository Information

Code for data cleaning and analysis is provided as part of the replication package. It is available at <http://www.openicpsr.org/openicpsr/workspace?goToPath=/openicpsr/183163>.

Dataset list

Below are a list of datasets used in the paper. The list includes raw data files and processed intermediate files. These are indicated in the “Notes” in the table below. The results from the archive can produced from the included processed intermediate files. Replication from raw data requires 4 files that cannot be redistributed to to licensing restrictions from the vendor. These are indicated as “No” in the “Provided” column. *Sample versions of these 4 files are included so the data fields and types can be replicated from the vendors.*

Data file	Source	Notes	Provided
Files in ../input			
<code>irs_cnty_8916_raw.sas</code>	IRS	Raw Data	Yes
<code>Incyallagi.sas</code>	IRS	Raw Data	Yes
<code>zip_agi0409.sas</code>	IRS	Raw Data	Yes
<code>Allhlcn.sas</code>	BLS	Raw Data	Yes
<code>qcew_8820.sas</code>	BLS	Raw Data	Yes
<code>1990.annual.singlefile.csv</code>	BLS	Raw Data	Yes
...			
<code>2019.annual.singlefile.csv</code>	BLS	Raw Data	Yes
<code>county_income.sas</code>	BEA	Raw Data	Yes
<code>cnty_unem.sas</code>	BLS	Raw Data	Yes
<code>stch1990.sas-stch1999.sas</code>	Census	Raw Data	Yes
<code>coest00intalldata.sas</code>	Census	Raw Data	Yes
<code>C-EST2020-ALLDATA.sas</code>	Census	Raw Data	Yes
<code>urban1990_CDC.xlsx</code>	CDC	Raw Data	Yes
<code>PctUrbalRural_County_2010.xlsx</code>	Census	Raw Data	Yes
<code>Education_attainment_usda_90_20_145yearACS.xlsx</code>	USDA	Raw Data	Yes
<code>Education_attainment_ACS_2010_5year_estimate.xlsx</code>	Census	Raw Data	Yes
<code>quarterly_real_GDP_growth.sas(GDPGROWTH.XLS)</code>	FRED	Raw Data	Yes
<code>wage_aggregate.sas (WAGES.XLS)</code>	FRED	Raw Data	Yes
<code>unrate_aggregate_monthly.sas (UNRATE.XLS)</code>	FRED	Raw Data	Yes
<code>fed_fund.sas (FEDFUNDS.XLS)</code>	FRED	Raw Data	Yes
<code>credit_spread_moody.sas (AAA.XLS and BBB.XLS)</code>	FRED	Raw Data	Yes
<code>funda_head.sas</code>	Compustat	Raw License Restricted	No
<code>hq_10k.sas</code>	SEC	Raw Data	Yes
<code>msi.sas</code>	CRSP	Raw License Restricted	No
<code>dsi.sas</code>	CRSP	Raw License Restricted	No
<code>cq_election.xlsx</code>	CQ	Raw License Restricted	No
<code>2002_NAICS_to_1987_SIC.xlsx</code>	Census	CrossWalk	Yes
<code>zip_county.sas</code>	HUD	CrossWalk	Yes
<code>state_ab.sas</code>	USPS	CrossWalk (state abbrev.)	Yes

Data file	Source	Notes	Provided
cnty_head_count.dta	—	Processed Intermediate Appendix Table	Yes
macro_vars_all.dta	—	Processed Intermediate Table 3 SEs	Yes
Files in ../output sample.dta	—	Final Dataset Fig. 1, Tables 1,2, &3	Yes
vote_cny.sas	—	Processed Intermediate input to sample.dta	Yes
msi_real.dta(.sas)	—	Processed Returns Table 3 SEs.	Yes
cnty_head_count.dta	—	Processed Intermediate Appendix Table	Yes
cnty_age_race.sas	—	Processed Intermediate input to sample.dta	Yes
cnty_ret_ind_w.sas	—	Processed Intermediate input to sample.dta	Yes
dem_share.dta	—	Processed Intermediate input to sample.dta	Yes
div_ratio_excltop.dta	—	Processed Intermediate Appendix Table	Yes
irs_cnty_8916.sas	—	Processed Intermediate input to sample.dta	Yes
merged_data.dta	—	Processed Intermediate input to sample.dta	Yes
state_return.sas	—	Processed Intermediate input to sample.dta	Yes
part_county0417.dta	—	Processed Intermediate Appendix Table	Yes
election_ret_m.dta	—	Processed Intermediate Appendix Table	Yes
edu_urban_90_10	—	Processed Intermediate input to sample.dta	Yes
orth_bs_true.dta	—	Processed Intermediate for Table 3 SEs	Yes

Computational requirements

Software Requirements

- Stata (code was last run with version 17) with the following .ado files installed:
 - `reghdfe`
 - `gsort`
 - `eststo`
 - `winsor2`
 - `ivreghdfe` (needed for appendix tables)
 - `ivreg2` (needed for ivreghdfe)

- `ranktest` (needed for `ivreghdfe`)
- `rangestat`
- `ftools`

These .ado files can be installed automatically using the included `config.do` file. Alternatively, they can be found using “`findit [command name]`” in stata and clicking on the link or “SSC install [command name]” and following the prompt.

- SAS 9.4 (code was run on 9.4 m6. If running on a version earlier than 9.4m4 `proc import/export` may change variable names to lower case. This may require minor code edits to STATA code to reflect lower case variable names).

Controlled Randomness

The bootstrapped standard errors in Table 3 use a random number seed set at lines 246 and line 433 of program 02 - `variable_construction_and_BS_SEs.do`. The specification curve produced for the Internet Appendix uses random number seeds set at lines 363, 601, and 616 in `programs/03 - appendix_tables_figures.do`.

Memory and Runtime Requirements

Summary Approximate time needed to reproduce the analyses on a standard (CURRENT YEAR) desktop machine: For all except bootstrapped standard errors for Table 3, 10-60 hours. For Bootstrapped SE’s for table 3, 8-24 hours on standard desktop computer running Stata 17.

Details The code was last run on a **Intel Core i7-10750H based laptop with Windows version 10**.

Boostrapped SEs for Table 3 were run on Intel i9 desktop, 3GHz with 18 cores, 128GB RAM and Windows 10.

Description of programs/code

The archive contains 2 SAS programs and 2 Stata programs and an excel file. The SAS programs process and merge data from the various data sources described above. The first Stata program produces the output contained in the tables and figures of the paper. The second stata file and the excel file produce the tables and figures contained in the internet appendix.

- `programs/01 - data_construction.sas`:
 - processes raw election data from `cq_election.xlsx`
 - processes raw IRS data from the IRS
 - merges and processes county level demographic data from sources described above
 - calculates returns based on companies located in the county. Used in the Internet Appendix. REQUIRES WRDS ACCESS (DISCUSSED ABOVE). ONLY NECESSARY FOR APPENDIX. PROCESSED DATA INCLUDED FOR REPLICATION IF NO WRDS ACCESS
 - merges corporate headquarters location. Used in the Internet Appendix.
 - creates the main dataset used in the analysis by merging various county level datasets together
 - used to calculate the different participation measures for the robustness tests presented in the Internet Appendix
 - used to calculate the different return measures for the robustness tests presented in the Internet Appendix
- `programs/02 - Paper_exhibits_BS_SEs.do`
 - constructs final variables for analysis
 - produces all regressions and estimates presented in tables and figures in the paper
 - calculates bootstrapped SEs for Table 3
- `programs/03 - appendix_tables_figures.do`
 - creates all exhibits from the Internet Appendix except the visualization of the specification curve (Internet Appendix Figures III and IV).
- `programs/spec_curve_visualization.xlsx`

- produces visualization of the specification curve (Internet Appendix Figures III and IV). Underlying data produced in `programs/03 - appendix_tables_figures.do`

Instructions to Replicators

For full replication from scratch, all source data must be placed in `../Input` directory. All source data save for those restricted by license are included in the archive.

- Edit all programs (01-03) to adjust the default path at the top.
- Install all necessary STATA .ado files as listed above using `config.do`, `"findit"`, or `"SSC install"`.
- Download all data not included in the repository and place in `../Input` directory.
- Run each program in sequence.

Processed intermediate files are included in the archive. All results except the bootstrapped standard errors from Table 3 can be produced directly from the archive by running only programs 02 and 03. Program 02 will reproduce the standard errors from Table 3 but intermediate bootstrapped samples are not included in the archive and must be produced from the program.

List of tables and programs

The provided code reproduces:

- X All numbers provided in text in the paper from included processed data
- X All tables and figures in the paper from included processed data

Note: Full replication from source data requires access and acquisition of CQ voting data, CRSP return data, the Compustat headquarters data as discussed above.

Figure/Table #	Program	Line Numbers/ Note
Figure 1	<code>programs/02 - Paper_exhibits_BS_SEs</code>	621-655
Table 1	<code>programs/02 - Paper_exhibits_BS_SEs</code>	119-122
Table 2	<code>programs/02 - Paper_exhibits_BS_SEs</code>	125-185
Table 3 Coefficients	<code>programs/02 - Paper_exhibits_BS_SEs</code>	538-569
Table 3 SEs	<code>programs/02 - Paper_exhibits_BS_SEs</code>	577-645
Appendix Exhibits	<code>programs/05 - appendix_tables_figures.do</code>	See Code Comments for Exhibit info
In-line 2008 counterfactual (McCain example, Sec 2.C)	<code>programs/02 - Paper_exhibits_BS_SEs</code>	194-217
In-line economic significance (multiple locations)	<code>programs/02 - Paper_exhibits_BS_SEs</code>	Can be calculated directly from numbers reported in Table 2 coefficients and Table 1 variable distributions.

References

- Board of Governors of the Federal Reserve System (2020). Federal funds effective rate [fedfunds]. Retrieved from FRED, Federal Reserve Bank of St. Louis; <https://fred.stlouisfed.org/series/FEDFUNDS> (accessed December 2022).
- Bureau of Economic Analysis (2020). Real gross domestic product [a191rl1q225sbea]. Retrieved from FRED, Federal Reserve Bank of St. Louis; <https://fred.stlouisfed.org/series/A191RL1Q225SBEA> (accessed December 2022).

Bureau of Economic Analysis (2022). Personal income by county, 1988-2020. Available on the BEA website, <https://www.bea.gov/itable/regional-gdp-and-personal-income>, Interactive Tables: Personal income by county and MSA (accessed September 2022).

Bureau of Labor Statistics (2020a). Consumer price index for all urban consumers: All items in u.s. city average [cpiaucsl]. Retrieved from FRED, Federal Reserve Bank of St. Louis; <https://fred.stlouisfed.org/series/CPIAUCSL> (accessed April 2021).

Bureau of Labor Statistics (2020b). Employed full time: Median usual weekly real earnings: Wage and salary workers: 16 years and over [les1252881600q]. Retrieved from FRED, Federal Reserve Bank of St. Louis; <https://fred.stlouisfed.org/series/LES1252881600Q> (accessed December 2022).

Bureau of Labor Statistics (2020c). Unemployment rate [unrate]. Retrieved from FRED, Federal Reserve Bank of St. Louis; <https://fred.stlouisfed.org/series/UNRATE> (accessed December 2022).

Bureau of Labor Statistics (2022a). Local area unemployment statistics, 1990-2020. Available on the BLS website, <https://www.bls.gov/lau/tables.htm> (accessed September 2022).

Bureau of Labor Statistics (2022b). Quarterly census of employment and wages, 1988-2020. Available on the BLS website, <https://www.bls.gov/cew/downloadable-data-files.htm> (accessed September 2022).

Census Bureau (1990,2010a). Educational attainment. Data in 1990 are obtained through the USDA <https://www.ers.usda.gov/data-products/county-level-data-sets/county-level-data-sets-download-data/> Dataset used: Educational attainment for adults age 25 and older for the U.S., States, and counties, 1970–2020 (accessed January 2021). Data in 2010 are obtained from the 2010 American Community Survey 5-year estimates, <https://data.census.gov/all?q=education> Under data series S1501 - 2010: ACS 5-Year Estimates Subject Tables; Geography: All counties within United States and Puerto Rico (accessed January 2021).

Census Bureau (1990,2010b). Urban and rural classification and urban area criteria. Data in 1990 are obtained through the CDC <https://www2.cdc.gov/nceh/lead/census90/house11/download.htm> Dataset used: cnty.dbf (accessed September 2017). Data in 2010 are obtained from the Census website, <https://www.census.gov/programs-surveys/geography/guidance/geo-areas/urban-rural/2010-urban-rural.html> Dataset used: Percent Urban and Rural In 2010 by State and County (accessed January 2021).

Census Bureau (2022a). Census naics and sic codes concordances. <https://www.census.gov/naics/?68967> Dataset used: 2002 NAICS to 1987 SIC (Accessed December 2022).

Census Bureau (2022b). County population by characteristics, 1990-2020. Data from 1990 to 2010 are obtained from the NBER <https://www.nber.org/research/data/us-intercensal-county-population-data-age-sex-race-and-hispanic-origin>. (accessed May 2017). Data from 2011 to 2020 are obtained from the Census website, <https://www2.census.gov/programs-surveys/popest/datasets/2010-2020/counties/asrh/> Dataset used: CC-EST2020-ALLDATA.csv (accessed September 2022).

Center for Research in Security Prices (2021). Crsp indexes (daily & monthly), 1988-2020. Technical report. Available to the University of Pittsburgh through Wharton Research Data Services (WRDS), <https://wrds-www.wharton.upenn.edu/pages/about/data-vendors/center-for-research-in-security-prices-crsp/> (accessed May 2021).

CQ Press (2022). Voting and elections collection. <https://library.cqpress.com/elections/> Presidential election years : 1984-2020 (Accessed December 2022).

HUD (2022). Usps zip code crosswalk files. https://www.huduser.gov/portal/datasets/usps_crosswalk.html Data year and quarter: 2010 Q1 (Accessed June 2020).

IRS (2021). Individual income tax zip code data. Technical report. Available on the IRS website <https://www.irs.gov/statistics/soi-tax-stats-individual-income-tax-statistics-zip-code-data-soi> (accessed April 2021).

- Moody's (2020). Moody's seasoned baa corporate bond yield [baa aaa]. Retrieved from FRED, Federal Reserve Bank of St. Louis; <https://fred.stlouisfed.org/series/BAA> and <https://fred.stlouisfed.org/series/AAA> (accessed December 2022).
- SEC (2020). Sec edgar database, 1988-2020. Technical report. Access via Company Search feature on SEC EDGAR server <https://www.sec.gov/edgar/searchedgar/companysearch> (accessed December 2020).
- S&P Global (2020). Compustat, 1988-2020. Technical report. Available to the University of Pittsburgh through Wharton Research Data Services (WRDS), <https://wrds-www.wharton.upenn.edu/pages/get-data/compustat-capital-iq-standard-poors/> (accessed December 2020).